# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Fourth Semester of B. Tech. (EC) Examination**

**April 2013**

**EC209 Microprocessors & Microcontroller**

**Date: 23.04.2013, Tuesday** **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

***Instructions:***

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

## SECTION – I

**Q - 1 (a)** List the four control signals commonly used by the 8085 MPU. **[03]**

**(b)** Fill in the blanks: **[04]**

(i) _______address lines are required to interface 4 GB memory.

(ii) ________is non maskable interrupt of 8085.

(iii) On reset of 8085, content of PC is _________.

(iv) In SRAM, data bit is store in terms of_________ .

**Q - 2 (a)** Define the term MPU and give the difference between Microprocessor & Microcontroller. **[06]**

**(b)** Draw and explain the timing diagram for opcode fetch machine cycle. **[04]**

**(c)** Write 8085 assembly program to convert Hex number located at 4150H memory location into its equivalent BCD number and store it at 4151H and 4152H memory location. **[04]**

**OR**

**Q - 2 (a)** Explain SIM and RIM instructions. **[06]**

**(b)** Write 8085 assembly program to check whether the fourth bit of a byte at location 2000H is a 0 or 1. If 0 store 00H at location 3000H, else store FFH at 3000H. **[04]**

**(c)** What is a flag? With help of example explain each flag. **[04]**

**Q - 3 (a)** With a neat diagram explain the 8085 based interfacing circuit using 74138 needed to connect the following **[07]**

(i) 2K × 8 RAM chip in the address space C000H-C7FFH

(ii) 4K × 8 RAM chip in the address space D000H-DFFFH

(iii) 8K × 8 EPROM chip in the address space 2000H-3FFFH

**(b)** Write 8085 assembly program to sort the data array of 16 bytes at memory location from 2010H to 201FH, in ascending form. **[04]**

(c) Define: Machine cycle and Instruction cycle. [03]

OR

Q - 3 (a) Discuss the programming model of 8085 microprocessor with the help of suitable diagram. [07]

(b) Write 8085 assembly program to generate delay of 0.5 second. [04]

(c) Differentiate maskable and non-maskable interrupts. [03]

## SECTION – II

Q - 4 (a) With suitable examples, explain Program Status Word (PSW) register of 8051. [04]

(b) List the logical instructions of 8085 microprocessor. [03]

Q - 5 (a) Draw the block diagram of 8255 and explain its mode and control word. [06]

(b) Explain different types of Jump instructions of 8051 with suitable examples. [04]

(c) Write 8051 assembly program to store the upper nibble of R0 in both nibbles of Register R7. For example, if R0=A6H then R7=AAH. [04]

OR

Q - 5 (a) Draw pin diagram of 8051 microcontroller and explain function of following pin: [06]

(i) EA, (ii) ALE

(b) Write 8051 assembly program to copy program (code) bytes 0100H to 0102H to internal RAM locations 20H to 22H. [04]

(c) List out difference between 'memory mapped I/O' and 'I/O mapped I/O'. [04]

Q - 6 (a) Explain the architecture of 8051 microcontroller with suitable block diagram. [06]

(b) Explain different addressing modes of 8051 in detail. [04]

(c) Write 8051 assembly program to add the number 84H to RAM locations 17H and 18H. [04]

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